



**Directorate of Information Technology
Govt. of Puducherry**



Examination Marks Software

Detailed Project Report (DPR)

Prepared by



Puducherry e-Governance Society (PeGS)

**Version 1.1
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Revision History

Version	Date	Changes
1.0	10-09-2025	Release of Version 1.0

Table of Contents

1. Introduction.....	4
2. Existing Software Application.....	4
3. Gap Analysis.....	5
4. Requirements of Examination Marks Software (EMS).....	5
4.1 Student Module	6
4.1.1 Basic Details.....	6
4.1.2 Education Details	7
4.1.3 Data Verification	7
4.2 Examination Module	8
4.2.2 Examination Schedule	9
4.2.3 Examination Attendance	10
4.3 Marks Module	10
4.3.1 Entry of Subject Marks	10
4.3.2 Entry of Student Marks.....	11
4.3.3 Generation of Marks Sheet	11
4.4 Admin Module.....	12
4.4.1 User Creation.....	12
4.5.2 User Roles	13
4.5.3 User Mapping	14
4.5.4 Feature Configuration	14
4.5 MIS Module.....	15
4.5.1 Student Reports.....	15
4.5.2 Examination Reports	15
4.5.3 Marks Reports.....	16
4.6 Data Migration	16
4.7.1 Consolidation of Meta Data	16
4.7.2 Data Cleaning	16
4.7.3 Data Import	17
4.7.4 Data Integration	17
5. Project Costing	18
6. Project Timelines.....	22

1. Introduction

The Board of Medical Education (BOME) and the Board of Examination in Nursing (BOEN) have the mandate to enrol candidate, conduct examinations and issue marks sheet for various diploma courses in medicine and nursing. The existing software was developed as stand-alone software on client-server architecture and accessible through local area network only. The software is being used for student data entry, marks entry and generation of marks sheet. Since the software has become outdated and incompatible with latest technologies, the Board intends to revamp the software with updated technologies. This document details the functional and technical requirements involved in redeveloping the software since it may not be possible to revamp the existing software.

2. Existing Software Application

The existing application comprises of features as shown in Figure-1.

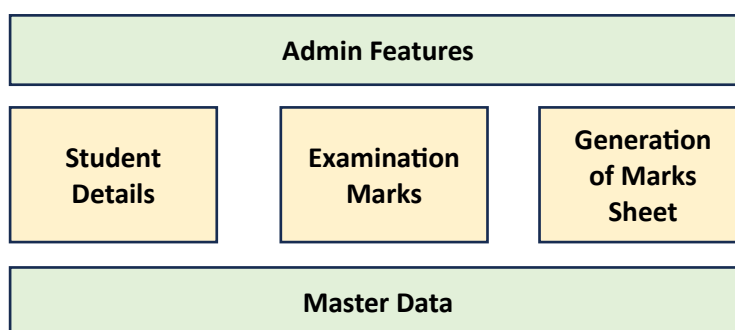


Figure-1: Features of Existing Software Application

The brief overview of each feature is as follows.

Table-1: Overview of Features

Module	Features
Admin Feature	Addition of Maser data Activation / Deactivation of Users Enabling / Disabling of options Correction of data
Student Details	Entry of student details Editing of student details
Examination Marks	Entry of Examination marks Editing of Examination marks
Generation of Marks Sheet	Generation of marks sheet in excel and pdf Printing of marks sheet
Master Data	Maintenance of master data

3. Gap Analysis

Some of the major gaps in features and functionalities have been identified and mentioned below.

- System has been developed on client-server technology and can be accessed only on local area network.
- System allows centralized data entry only.
- System cannot be integrated with external systems due to obsolescence of technology and incapability to be accessed on internet.
- System does not have workflow systems for hierarchical movement of tasks and data.
- System mandates saving marks sheet in PDF that should be printed in pre-printed marks sheet

4. Requirements of Examination Marks Software (EMS)

The new system (Examination Marks Software – EMS), addressing the procedural and functional gaps along with new technical and functional features, is proposed to have the following modules.

1. Student Module
2. Examination Module
3. Marks Module
4. Admin Module
5. MIS Module

The proposed modular architecture is proposed as follows.

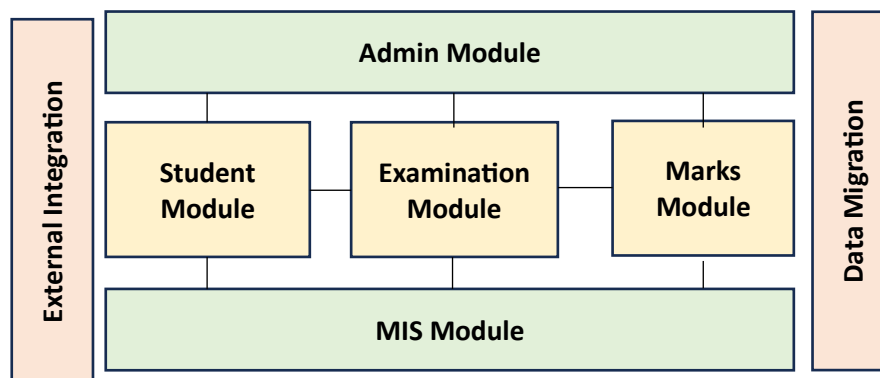


Figure-2: Proposed Modular Architecture

Core Modules:

- Student Module
- Examination Module
- Marks Module

Supporting Modules

- Admin Module
- MIS Module

The functional requirements of each module are elaborated as follows.

4.1 Student Module

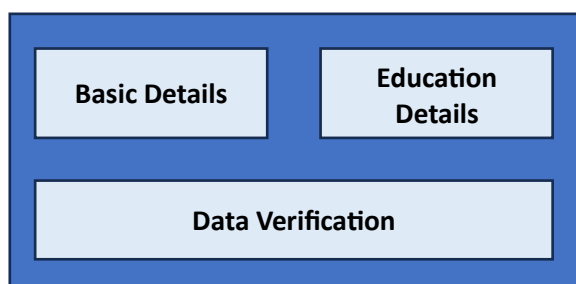


Figure-3: Sub-Modules of Student Module

The functional requirements of each sub-module are elaborated below.

4.1.1 Basic Details

Ref. No.	Requirement	Details
1	Entry of basic details of student	Provision to be given to enter the following basic details of students by respective colleges. - Name of Student - Date of Birth - E-Mail ID - Mobile Number - Student ID - Name of Father - Name of Mother - Address - Place of Residence - State - Pin Code System should allow upload of student details in the form of excel sheet. Provision to be given for colleges to download excel sheet template for student details. Verification of mobile number and E-Mail ID need to be provisioned in the form of clickable links.
2	Upload of documents	Provision to be made to upload documents of students for verifying the following. - Name of Student - Date of Birth

Ref. No.	Requirement	Details
		- Student ID - Address

4.1.2 Education Details

Ref. No.	Requirement	Details
3	Entry of education details of students	Colleges should be able to enter the following educational details of students. - Education Level (X, XII, Others) - Major Subjects - Name of Board / Exam - Name of School / College - Address of School / College - Place - Pin Code - Year of Passing - Total Marks - Marks Obtained - Grade / Percentage - Student Registration Number - Certificate Serial Number System should allow upload of education details in the form of pre-filled excel sheet. Provision to be given for colleges to download excel sheet template for entering education details.
4	Upload of documents for educational details	Colleges need to upload documents against each education detail in the form of PDF document. System should make it mandatory to upload documents against each education detail.

4.1.3 Data Verification

Ref. No.	Requirement	Details
5	Verification and approval of student details.	Provision to be given for verifying the data entered / uploaded by colleges with regard to basic details and education details. System should allow maker-checker-approver workflow for verifying the details at checker level and approving the details at approver level. Verifying and Approving need to be performed at Board level (centralized operation).

Ref. No.	Requirement	Details
		Approval of student details need to be done using digital signature certificate of competent official. Provision to be given to approve student details in the form of a list instead of approving each student at a time. System should be able to digitally sign the details of each student though the approval is done in the form of a list. The officials at Board level should be able to reject the student details or send back student details for modification in data entry and / or upload of correct documents. System should not allow modification of data at verification and approval level. Any correction / modification need to be performed at college level.
6	Generation of Unique Student Registration Number	System should be able to generate unique student registration number which will be used as student identity number for appearing for examination and entry of examination marks and generation of marks sheet.

4.2 Examination Module

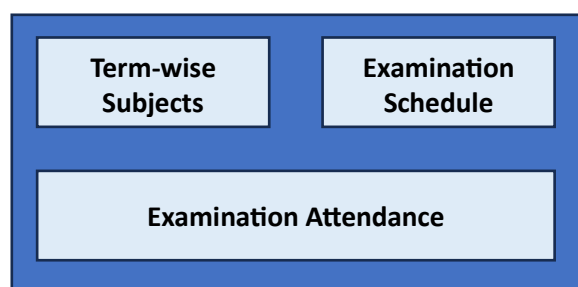


Figure-4: Sub-Modules of Examination Module

The functional requirements of each sub-module are elaborated below.

4.2.1 Examination Subjects

Ref. No.	Requirement	Details
7	Selection of subjects for each Term	Colleges should be able to select subjects pertaining to each Term of courses being offered.

Ref. No.	Requirement	Details
		System should allow adding / removing subjects from the already available subjects within each Term. Provision to be given to mention “effective start date” in case of addition of subjects and “effective end date” in case of removal of subjects.
8	Creation of new subjects / mapping of existing subjects	System should allow Admin to create new subjects and map the newly created subjects to courses with “effective start date” Provision to be given to map the existing subjects to new courses / existing courses with “effective start date” All existing subjects mapping to each course should have a default “effective start date” as decided by colleges.

4.2.2 Examination Schedule

Ref. No.	Requirement	Details
9	Creation of Examination Schedule for each Course and each Term	Colleges should be able to create Examination Schedule for each Course and for each Term indicating the date, time and venue of Examination for each subject. System should allow creation of mid-Term Examination and end-Term Examination with date, time and venue for each subject. Provision to be given to mark holidays and weekends during creation of Examination Schedule. Also, system should be able to generate alerts when Exams are being scheduled on general holidays and reserved holidays. Examination schedule need to be verified and approved at college level prior to publishing it on the portal.
10	Availability of Examination Schedule on portal	Provision to be given for students to download Examination Schedule from portal on searching the college, course and term. The latest Examination Schedule need to be made available for download with automatic disabling of pervious Examination Schedule after completion of Examinations.

4.2.3 Examination Attendance

Ref. No.	Requirement	Details
11	Marking of examination attendance for each student	System should allow marking of attendance of students who had appeared for examination. Provision to be given to upload attendance sheet for each subject. The document should be uploaded in PDF format.
12	Editing of examination attendance	System should allow competent officials only to edit the attendance data that has been saved and confirmed in the system. System should keep trail of such corrections in examination attendance data.

4.3 Marks Module

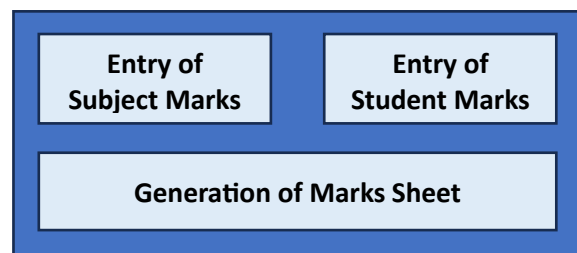


Figure-5: Sub-Modules of Marks Module

The functional requirements of each sub-module are elaborated below.

4.3.1 Entry of Subject Marks

Ref. No.	Requirement	Details
13	Entry of subject marks including pass mark	Provision to be given to enter total marks and pass marks for each subject. System should allow division of marks into internal and examination and specifying total marks for each division that add up to total marks. Provision to be given to specify pass mark either for each division or for the subject as a whole. The entry of subject marks should be specified with “effective start date”
14	Change of subject marks including pass mark	System should allow for change in subject mark including pass mark. Such changes need to be verified and approved by competent officials.

Ref. No.	Requirement	Details
		Any change in subject marks should be specified with “effective start date”. Changes to subject marks need to be digitally signed by competent official. System should keep trail of changes done on subject marks.

4.3.2 Entry of Student Marks

Ref. No.	Requirement	Details
15	Entry of student marks against each subject of a Term	System should allow colleges to enter marks of students for the examination attended by them. If a student is absent for an examination, then, system should not allow for entry of marks for the particular student and for that Examination and for that subject. System should allow entry of division-wise marks for each student. On entering the marks, system should be able to indicate “Pass” or “Fail” depending on the passing criteria. The marks entered for each student need to be verified and approved by competent officials. The approval of student marks need to be digitally signed by competent official
16	Change in entered student marks	Any change in already approved student marks should be done only by competent official with digital signature. System should keep trail of all changes to student marks done by competent officials and should be readily available at any point of time.

4.3.3 Generation of Marks Sheet

Ref. No.	Requirement	Details
17	Generation of Marks Sheet after approval of marks	System should be able to generate Marks Sheet in the format as decided by department. Provision to be made to generated Marks Sheet in customized format so as to avoid use of pre-printed Maek sheet.

Ref. No.	Requirement	Details
		System should generate unique reference number for each Marks Sheet. Further, the Marks Sheet should display registration number of the student that was assigned during entry of student details. The following rule need to be adopted for marking divisions of marks. >80% Distinction 60-80%: First Class 45-50%: Second Class 35-45%: Third Class <35%: Fail
18	Digitally signed Marks Sheet	The Marks Sheet need to be digitally signed with digital signatures embedded in the Marks Sheet. The digital signature of the competent official in the Board should be displayed on the Marks Sheet.
19	Integration with Digilocker	Marks Sheet need to be integrated with Digilocker to allow students to access Marks Sheet through Digilocker platform.

4.4 Admin Module

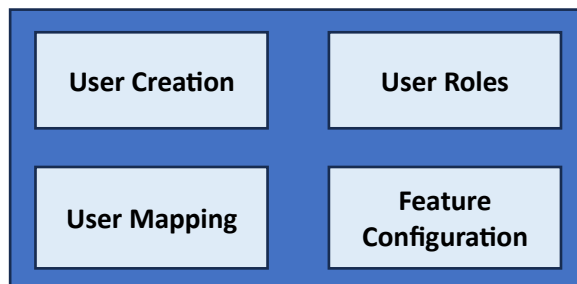


Figure-6: Sub Modules of Admin Module

The functional requirements of each sub-module are elaborated below

4.4.1 User Creation

Ref. No.	Requirement	Details
20	Creation of Users	Provision to create login credentials for staff of colleges and officials of Boards at individual level. The following details of staff / officials need to be entered / uploaded into the system. - Name of Staff / Official

Ref. No.	Requirement	Details
		- Name of College / Board - Designation - Mobile Number - E-Mail ID - Address - Place - Pin Code System should allow OTP based authentication for logging into the portal. Provision to be given to change mobile number and e-mail ID..
21	Deactivation / Activation of Users	System should have provision to deactivate the user on account of the following. - resignation from services - suspension - superannuation - transfer to other work domain System should allow temporary deactivation / activation of users on need basis.

4.5.2 User Roles

Ref. No.	Requirement	Details
22	Assigning of Roles to Users	System should allow Admin to assign roles to users based on tasks carried out by users. The roles are categorized as per the following. - Admin Roles - Staff/ Official Roles - Feature Roles - Reports Roles Admin Roles: These roles when assigned to users would allow creation / modification / deactivation / activation of users in the system. These roles will also allow changes to system configurations and task movement. Staff / Official Roles: These roles allow viewing of tasks and performing activities on tasks. Cration, verification and approval are the primary tasks performed by staff and officials.

Ref. No.	Requirement	Details
		Feature Roles: These roles provide the users to perform various feature roles pertaining to student module, examination module and marks module. Reports Roles: These roles allow users to view and download reports from the system. It also allows to submit requirements for provision of new reports that may be useful in decision making. “Super Admin” role will allow the user to control all the roles and to monitor the users assigned with critical roles.

4.5.3 User Mapping

Ref. No.	Requirement	Details
23	Mapping of hierarchies for task workflows	System should allow mapping of office locations to higher officials to allow for movement of tasks. Further, provision to be given to map offices across colleges and Boards to facilitate inter-organizational work flows. The mapping need to be performed on offices and not on individual users. It is necessary to keep the master data of offices without options to edit / delete the records.
24	Modification in hierarchy mapping	System should allow modification in mapping of hierarchies. In this process, some office locations may be removed from the task workflow and some may be added. System should verify the availability of tasks in to the user mapped to an office location during removal of office location from the hierarchy. In such case, system should not allow for the office location to be removed till the tasks are completed.

4.5.4 Feature Configuration

Ref. No.	Requirement	Details
25	Configuration of system for enabling / disabling features	System should allow Super Admin to enable / disable features based on specific requirements of departments.

Ref. No.	Requirement	Details
		These include task level configuration and work flow configurations.
26	Incorporation of new features	Provision for Super Admin to release new features incorporated in the system. The new releases need to be authenticated by Super Admin prior to opening the features for users.

4.5 MIS Module

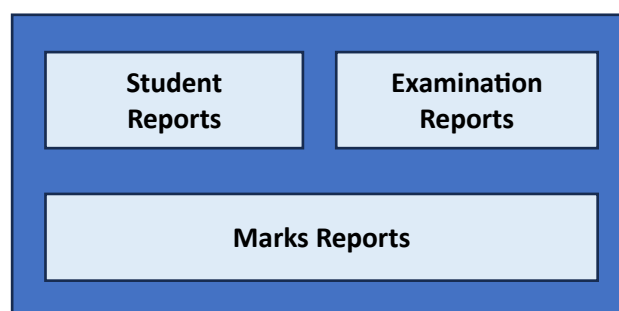


Figure-7: Sub Modules of Reports Module

The functional requirements of each sub-module are elaborated below

4.5.1 Student Reports

Ref. No.	Requirement	Details
27	Generation of Student Reports	Provision to generate the following indicative student reports - College-wise enrolment of students - Course-wise enrolment of students - Age-wise distribution of students - Region-wise distribution of students - Year-wise enrolment of students

4.5.2 Examination Reports

Ref. No.	Requirement	Details
28	Generation of Examination Reports	Provision to generate the following indicative examination reports - College-wise Examination Report - Course-wise Examination Report - Examination Attendance Report - Year-wise Exam Attendance Report - Exam Postponement Report

4.5.3 Marks Reports

Ref. No.	Requirement	Details
29	Generation of Marks Reports	Provision to generate the following indicative marks reports - College-wise Pass-Fail Report - College-wise Division Report - College-wise Marks Distribution Report - Student-wise Marks Trend Report - Course-wise Pass-Fail Report - Course-wise Division Report - Course-wise Marks Distribution Report

4.6 Data Migration

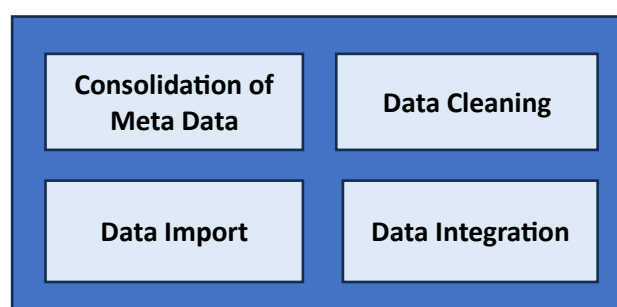


Figure-8: Tasks of Data Migration

The tasks under data migration are elaborated below

4.7.1 Consolidation of Meta Data

Ref. No.	Requirement	Details
30	Preparation of meta data of existing database structures	The existing database structure and tables need to be studied with regard to meta data, constraints and validations. The relevant meta data extracted from database tables need to be consolidated along with data constraints and data validations.
31	Creation of database tables based on meta data	Database tables need to be created based on consolidated meta data to facilitate import of data.

4.7.2 Data Cleaning

Ref. No.	Requirement	Details
32	Cleaning of existing data available in database tables	Data available in existing database tables need to be cleaned based on the following criteria - Improper data format - Blank data records - Incomplete data

Ref. No.	Requirement	Details
		- Repetitive data Attempt to be made to clean data with tools and methods for facilitating saving of trail of data cleaning activity. The steps undertaken the data cleaning activity need to be documented and the results at each stage to be saved for audit of activity.

4.7.3 Data Import

Ref. No.	Requirement	Details
33	Import of clean data into database tables	Data obtained after data cleaning activity need to be imported into database tables that have been created from the consolidated meta data. Care should be taken to maintain the relations between tables along with data constraints and data validations. The imported data need to be saved as a separate database scheme without being part of main database of the new system.

4.7.4 Data Integration

Ref. No.	Requirement	Details
34	Integration of imported data into main database	The imported data need to be integrated into the main database. The activity need to be performed table by table, checking for data correctness after each integration. The results obtained after each integration need to be documented for subsequent reference and audit.

5. Project Costing

The following are the norms followed for calculating project cost.

Level of Complexity:

Level	Details
Low	Screens involving data entry, saving and data retrieving and MIS reports that query less than 3 tables
Medium	Screens involving validations, authentications, digital signing, OTP integrations, workflows and MIS reports that query more than 3 tables
High	Screens involving mathematical and logical calculations, external integrations, reconciliation and MIS reports that query more than 5 tables

Number of Man-days

Data Fields	Level	Man-days	Remarks
10-12 data fields	Low	8 days	Proportionate if less or more data fields
10-12 data fields	Medium	16 days	Proportionate if less or more data fields
10-12 data fields	High	24 days	Proportionate if less or more data fields
Task Submission	Low	2 days	
Task Submission	Medium	4 days	
Task Submission	High	6 days	

Distribution of Software Development Activity

Activity	Distribution
Requirement Study Preparation of System Requirement Specification Workflow documentation	20%
Design of Database Design of Screens Development of Application External and Internal Integrations	50%
Testing of Application Project Management Project Documentation User Acceptance Testing (UAT)	30%

Based on the above norms, the detailed worksheet for calculating the number of man-days required for the project is given below.

Sl. No.	Module	Sub Module	Requirement	Low	Med.	High	Data Fields	Man days
1	Student Module	Basic Details	Entry of basic details of student		Yes		11	16
2			Upload of documents		Yes			8
3		Education Details	Entry of education details of students			Yes	13	24
4			Upload of documents for educational details		Yes			8
5		Data Verification	Verification and approval of student details			Yes		24
6			Generation of Unique Student Registration Number	Yes				4
7	Examination Module	Examination Subjects	Selection of subjects for each Term		Yes		10-12	16
8			Creation of new subjects / mapping of existing subjects		Yes		8-10	10
9		Examination Schedule	Creation of Examination Schedule for each Course and each Term		Yes			20
10			Availability of Examination Schedule on portal	Yes				10
11		Examination Attendance	Marking of examination attendance for each student	Yes				8
12			Editing of examination attendance	Yes				6
13	Marks Module	Entry of Subject Marks	Entry of subject marks including pass mark		Yes		10-12	16

Sl. No.	Module	Sub Module	Requirement	Low	Med.	High	Data Fields	Man days
14			Change of subject marks including pass mark			Yes		20
15		Entry of Student Marks	Entry of student marks against each subject of a Term		Yes			20
16			Change in entered student marks	Yes				8
17		Generation of Marks Sheet	Generation of Marks Sheet after approval of marks		Yes			20
18			Digitally signed Marks Sheet			Yes		10
19			Integration with Digilocker			Yes		10
20	Admin Module	User Creation	Creation of Users		Yes		5	8
21			Deactivation / Activation of Users	Yes				2
22		User Roles	Assigning of Roles to Users		Yes		15	16
23		User Mapping	Mapping of hierarchies for task workflows	Yes				4
24			Modification in hierarchy mapping	Yes				4
25		Feature Configuration	Configuration of system for enabling / disabling features		Yes			10
26			Incorporation of new features		Yes			10
27	MIS Module	Student Reports	Generation of Student Reports		Yes			20
28		Examination Reports	Generation of Examination Reports		Yes			20
29		Marks Reports	Generation of Marks Reports			Yes		30
30	Data Migration	Consolidation of Meta Data	Preparation of meta data of		Yes			8

Sl. No.	Module	Sub Module	Requirement	Low	Med.	High	Data Fields	Man days
			existing database structures					
31			Creation of database tables based on meta data		Yes			6
32		Data Cleaning	Cleaning of existing data available in database tables			Yes		10
33		Data Import	Import of clean data into database tables		Yes			16
34		Data Integration	Integration of imported data into main database			Yes		20
								442

The activity-wise breakup of man-months (considering 22 days in a month) is given below.

Activity	Distribution	Man-days	Man-months	Cost (Rs. 60,000 pm)
Requirement Study	20%	88.4	4.02	2,41,090.91
Design and Development	50%	221	10.05	6,02,727.27
Testing and Development	30%	132.6	6.03	3,61,636.36
Software Development Cost				12,05,454.55
Software Maintenance Cost (10%) – 1 year				1,20,545.45
Total Software Cost				13,26,000.00
Servers – NICSI cloud (1 year)				2,00,000.00
Software Licenses and Audit				1,00,000.00
Project Cost				16,26,000.00
6% PeGS charges				97,560.00
18% GST				2,92,680.00
Total Cost				20,16,240.00

6. Project Timelines

With deployment of Business Analyst, full stack software developers and tester, the time-line for the project is as follows.

Month	1	2	3	4	5
Study (2 no.)					
Development (3 nos.)					
Testing & Deployment (2 no.)					

Expected Time-line: **5 months**

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